8.2 Run-Time Power Management System by On-Die Power Sensor with Silicon Machine Learning-Based Calibration in a 3nm Octa-Core CPU

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For flagship smartphones, the gaming experience has become one of essential demands, requiring high frame rate per second (fps), display quality and durability. These demands drive the evolution of the CPU to enhance performance and computing efficiency. The CPU offers multiple cores to support various computing tasks in an energy-efficient manner, relying on the scheduler [1] to allocate tasks to each core based on available power budgets. Power budgets are calculated at intervals of each computing thread. As games run at higher fps, the intervals of power budgeting become shorter. The average interval of power budgeting in two top-gaming applications is shown in Fig. 8.2.1. With fps rising to 90, the average interval is shortened to less than 50μs. Meanwhile, the CPU clock speeds increase to enhance performance for gaming, resulting in ever-higher current. Peak current grows faster than average current when high CPU performance is needed. The growing peak current causes over-current excursions to exceed the PMIC tolerance, resulting in system instability that degrades CPU performance. Therefore, a hardware-based power sensing system is essential to enable fast power/current management, for the purpose of achieving gaming performance, computing efficiency and sustained power delivery for the CPU.

In this work, a run-time power/current management system consisting of an on-die power sensor (ODPS) is demonstrated in an octa-core CPU. The CPU is part of the mobile 5G flagship SoC, implemented in 3nm FinFET technology. The ODPS calculates the power using a current sensor (I-sensor) and a synchronous voltage sensor (V-sensor). To enhance power sensing accuracy, a silicon-based AI/Machine Learning (ML) model is proposed for calibration. The ODPS achieves 94.83% CPU power accuracy for the GeekBenchv6 CPU benchmark, accounting for run-time voltage and temperature fluctuations. The ODPS occupies 0.0313% of the whole CPU area. Moreover, the ODPS is capable of monitoring CPU on-time current in 10μs sampling windows. When applied to over-current control (OCC), it limits the over-current excursion on the PMIC to ensure power stability without observable performance loss. When running a game, the ODPS profiles per-core CPU power consumption, providing dynamic run-time conditions to optimize power for a better user experience and battery life.

Figure 8.2.2 depicts the CPU comprising eight single-thread cores in the SoC, supporting the ARMv9.2 instruction set. The cores are categorized into three performance groups. The first gear includes an ARM Cortex-X925 High-Performance (HP) core clocked at up to 3.63GHz. The second gear features three ARM Cortex-X4 Balance-Performance (BP) cores at 3.3GHz, and the third gear comprises four 2.4GHz Cortex-A720 cores for High-Efficiency (HE) computing. To implement a power sensor, current and voltage readings are essential for power calculation. As shown in Fig. 8.2.3, the fully digital I-sensor [2] is implemented in each core to capture the run-time current. The I-sensor senses the current by measuring the voltage drop as delta speed counts (ΔRO) using two identical ring oscillators (ROSC). It converts ΔRO to current using a calibrated slope (counts vs. voltage) and Direct-Current Resistance (DCR). Since both slope and DCR vary with voltage and temperature, V-sensor and Temperature sensor (T-sensor) are necessary for accurate current calculation. The V-sensor provides voltage readings used not only for I-sensor hardware calibration, but also for power computation. The designated ROSC of I-sensor ($ROSC_I$) can detect μV-level voltage drops, and its speed counts (RO) are calibrated against a trim voltage step with small nonlinearity error, making $ROSC_I$ a reliable indicator of supply voltage.

In Fig. 8.2.3, the proposed V-sensor synchronously records the RO of $ROSC_I$ to quantify the run-time voltage after each sampling window. It uses the RO look-up-table with calibrated counts vs. voltage to find the closest RO_n and RO_{n+1} and their corresponding trim step voltages V_n and V_{n+1}. The run-time voltage (*V* in Fig. 8.2.3) is then interpolated between V_n and V_{n+1} based on the RO readings between RO_n and RO_{n+1}. The voltage *V* is used for both I-sensor slope/DCR calibration and ODPS power calculation. Leveraging the same trimming voltage step and calibration table as the I-sensor, the V-sensor achieves a 96.8% accuracy, ranging from 65% to 120% of the CPU supply voltage (V_{DD}). Using the T-sensor, a two-point temperature characterization is done [2]. Then, the I-sensor outputs ΔRO and converts it to current *I*. As shown in Fig. 8.2.3, this current conversion follows the formula of Counts-to-Ampere Multiplier (M_{CA}) based on ΔRO. The M_{CA} is the multiplicative inverse of the slope·DCR, which is a polynomial function of different process-sensor (P-sensor, i.e., $ROSC_I$ and $ROSC_{DCR^*}$) counts with varying coefficients (i.e., C_I and C_{DCR^*}), that align with the hardware design at specified step voltages (i.e., V_n and V_{n-1}). Both P-sensor counts and coefficients can vary with voltage to determine the M_{CA}.

In addition to the original terms of the M_{CA}, to enhance accuracy, several potential P-sensors terms are considered to identify more process-related correlations beyond the designated model. For example, wire variations between true and virtual V_{DD} in Fig. 8.2.3 can affect the I-sensor accuracy, and N/PMOS unbalances may also distort the ROSC counts at the worst corner. Thus, the M_{CA} model is extended to include extra P-sensor coefficient terms for regression based on AI/ML calibration, as illustrated in Fig. 8.2.4. The AI/ML calibration model leverages extensive silicon data to refine the M_{CA} for accuracy convergence. Taking the current measured by the oscilloscope as golden, the current I converted by M_{CA}, ΔRO (RO) from the I-sensor (P-sensors) are trained in various scenarios, such as generic benchmarks including Geekbenchv6, Antutu, and SpecInt2k6 etc. This training features good tolerances against voltage and temperature fluctuations. AI algorithms are used to elect the most relevant RO terms (i.e., P-sensors), thereby simplifying the coefficient terms of the model. The simplified M_{CA} model then utilizes ML method to efficiently train the coefficients for each elected RO term. Finally, the elected RO terms and their corresponding coefficients are precisely determined for each trim step voltage in one-shot effort. Figure 8.2.4 shows the per-die calibration example for the M_{CA} of the HP core whose V_{DD} range from 60% to 120%. Following this, the ODPS calculates the power by obtained voltage and current (ΔRO x M_{CA}).

Figure 8.2.5 shows the ODPS measured power and voltage when the CPU runs Geekbenchv6/SpecInt2k6 benchmarks. In the HP core, the run-time power measured by the ODPS agrees well (R^2=0.9602/0.952) with the oscilloscope measurements, achieving low inaccuracy within 5.17%/4.76%. The ODPS also profiles the run-time power of the three BP and four HE cores. It shows the on-time power core-to-core variations of up to ±25/16.7% at 102.5%/95% V_{DD} between the BP cores, and ±28% at 98.75% V_{DD} between the HE cores. Figure 8.2.6 illustrates the architecture of the ODPS-oriented power management system. When the CPU runs games, the ODPS provides power readings for profiling and budgeting in each core. These power readings can be transmitted in sequence from CPU to SoC/OS via a microprocessor (μP). Smartphone applications can ascertain the per-core breakdown in run-time power. This benefits power optimization, yielding a better user experience and battery life. Additionally, the ODPS monitors run-time peak current in 10μs sampling windows. Once the current exceeds a threshold, the OCC throttles the CPU clock speed to lower the current for protection. Silicon validation shows that the OCC limits the current according to PMIC tolerance and thus stabilizes CPU power delivery.

The die photo in Fig. 8.2.7 shows the multicore CPU with area of 21.42mm². This work demonstrates a run-time power management system in a 3nm CPU complex. The ODPS includes the fully digital I-sensor and V-sensor operating at the full range of the CPU supply voltages. The V-sensor leverages the same calibration procedure as the I-sensor to save characterization effort, achieving 96.8% voltage accuracy. For the I-sensor, an AI/ML calibration model is adopted to improve accuracy. Incorporated with the V-sensor, the ODPS achieves a power sensing accuracy of 94.83%, occupying 0.0313% of the total CPU area. In addition to offering per-core power for budgeting, the ODPS also monitors and controls the peak current in 10μs sampling windows as the CPU power/current management system, which stabilizes CPU power without observable performance penalties. The sequential CPU run-time power can be displayed and profiled by each core in smartphone applications, benefiting power optimization for better user experience and battery life.

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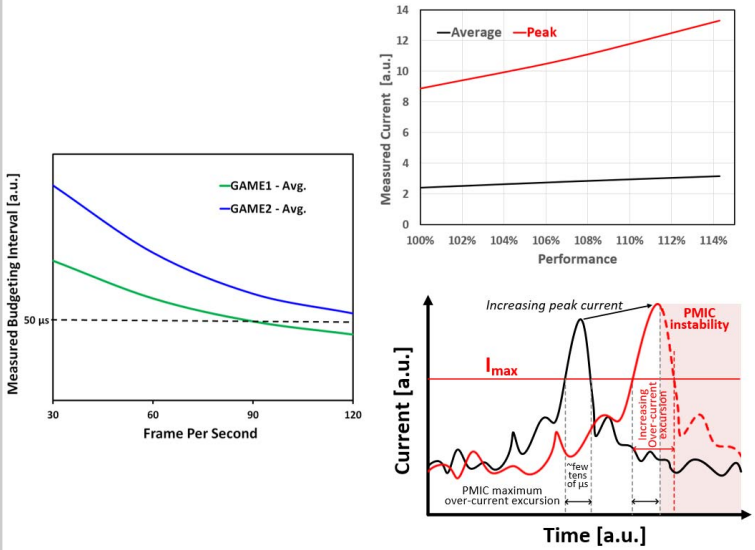

Figure 8.2.1: Ever-faster power-budgeting calculation and over-current excursion demands in 3nm gaming-centric CPU.

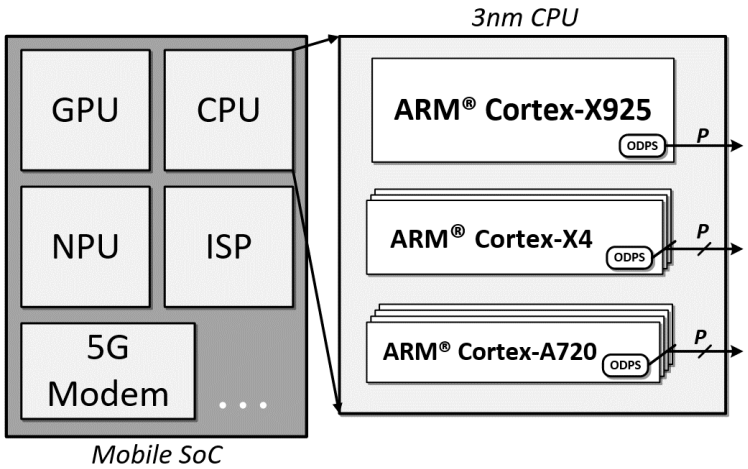

Figure 8.2.2: The 3nm CPU block diagram and architecture in SoC.

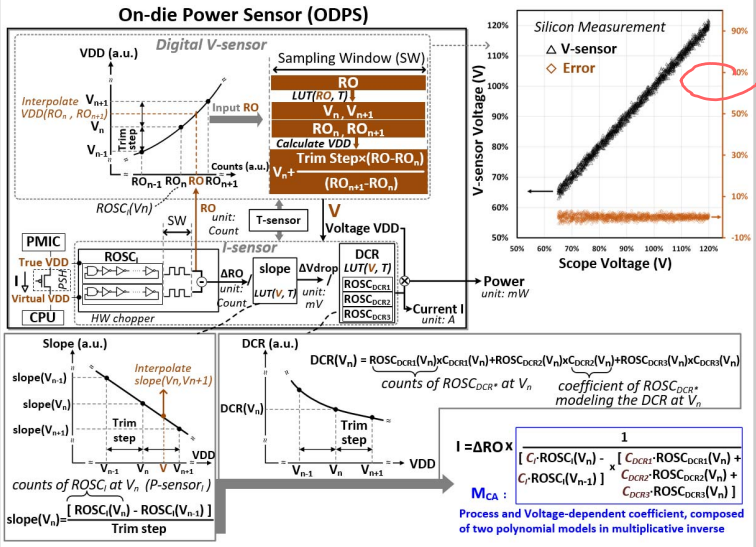

Figure 8.2.3: The proposed fully digital ODPS/V-sensor with calibration, and silicon measurements of the V-sensor.

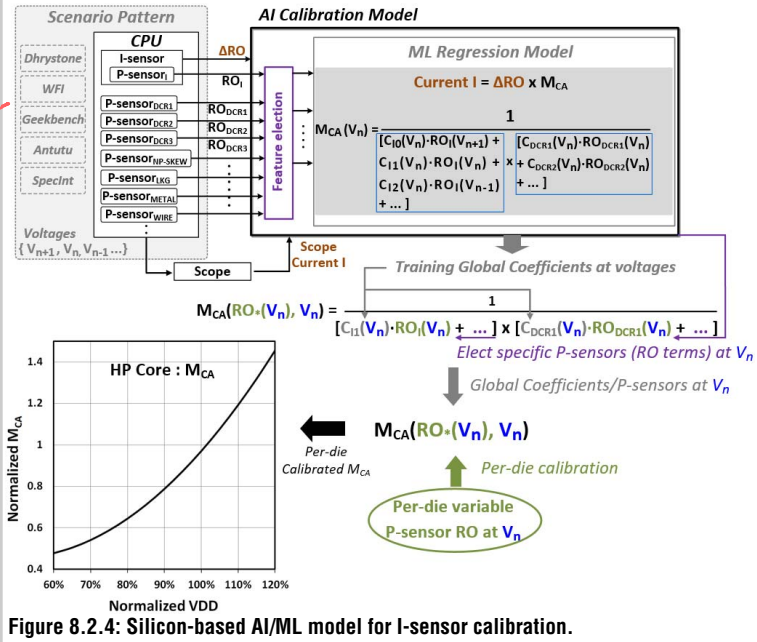

Figure 8.2.4: Silicon-based AI/ML model for I-sensor calibration.

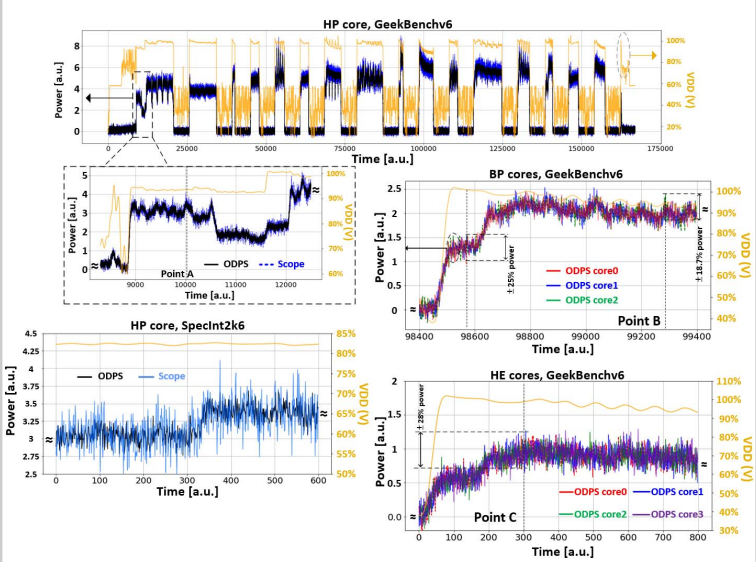

Figure 8.2.5: The ODPS measurement results of the HP core, BP cores and HE cores.

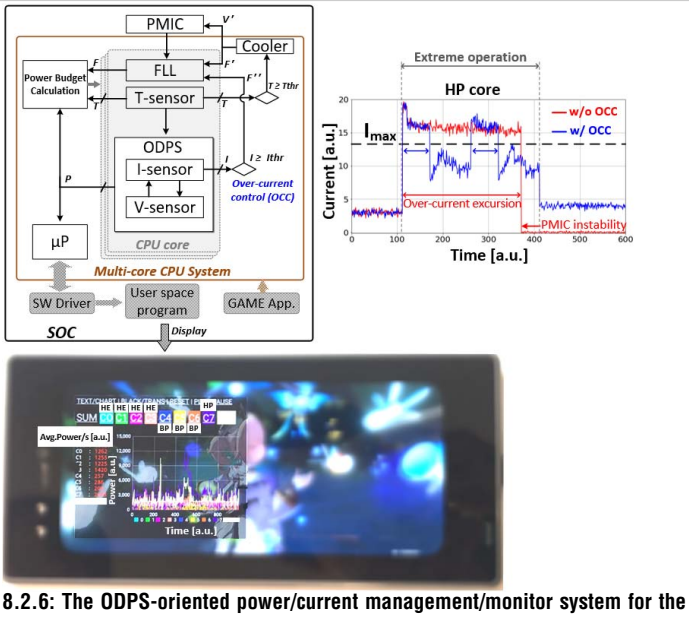

Figure 8.2.6: The ODPS-oriented power/current management/monitor system for the CPU with silicon results and on-phone demonstration.

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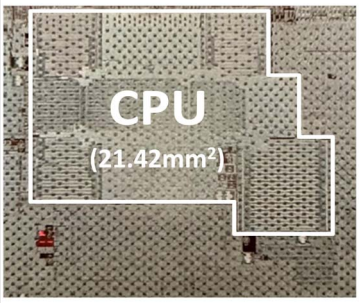

Feature	Digital Voltage Sensor	On-die Power Sensor
Technology	tsmc 3nm	
Area	6694.65um² (ODPS including V-sensor, ~0.0313% of CPU)	
Response time	10 µs (Configurable 1 ~ 1000 µs)	
Accuracy	96.8%	94.83%
Supply	65%~120% VDD of CPU	65%~105% VDD of CPU
Power	0.042% of HP core@ point A in Fig. 5 0.064% of BP cores@ point B in Fig. 5 0.156% of HE cores@ point C in Fig. 5	

Figure 8.2.7: The 3nm CPU die-photo and ODPS/digital V-sensor feature summary.

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